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Multimodal integration reveals complementary imaging and genetic predictors of coronary artery disease risk

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Take-home Cardiac imaging and polygenic risk capture complementary information about future CAD, improving risk stratification beyond clinical variables in UK Biobank.

Study design and imaging information

Clinical risk scores use a small set of traditional variables. We asked which imaging modalities add predictive information for incident coronary artery disease (CAD) once clinical, metabolic, electrocardiogram (ECG) and genetic risk are accounted for, and how imaging-derived and polygenic risk relate when measured in the same people.

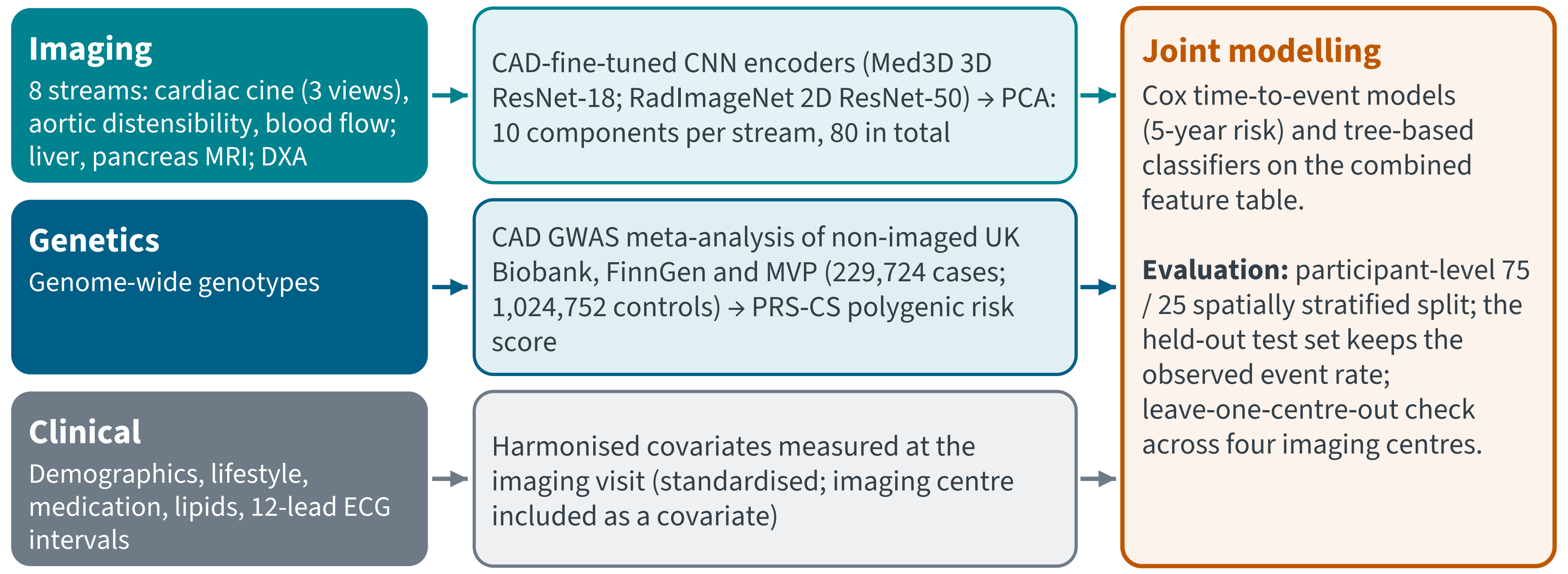
42,351
participants, UK Biobank imaging cohort

677
incident CAD cases (1.6% observed event proportion)

4.6 y
median follow-up; 5-year risk horizon

75 / 25
training 31,754 (499 events) / held-out test 10,597 (178 events)

Multimodal workflow



Imaging
8 streams: cardiac cine (3 views), aortic distensibility, blood flow; liver, pancreas MRI; DXA

CAD-fine-tuned CNN encoders (Med3D 3D ResNet-18; RadImageNet 2D ResNet-50) → PCA: 10 components per stream, 80 in total

Genetics
Genome-wide genotypes

CAD GWAS meta-analysis of non-imaged UK Biobank, FinnGen and MVP (229,724 cases; 1,024,752 controls) → PRS-CS polygenic risk score

Clinical
Demographics, lifestyle, medication, lipids, 12-lead ECG intervals

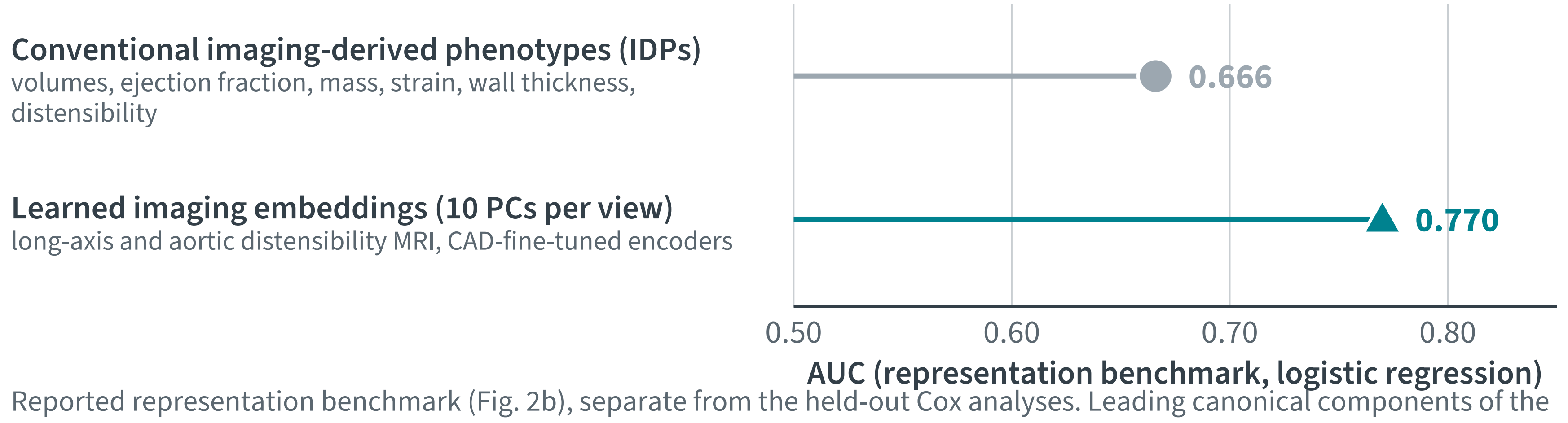
Harmonised covariates measured at the imaging visit (standardised; imaging centre included as a covariate)

Joint modelling
Cox time-to-event models (5-year risk) and tree-based classifiers on the combined feature table.

Evaluation: participant-level 75 / 25 spatially stratified split; the held-out test set keeps the observed event rate; leave-one-centre-out check across four imaging centres.

Outcome: clinically recognised CAD from linked hospital and death records (myocardial infarction, angina or ischaemic heart disease, coronary revascularisation); first event after the imaging visit, prevalent CAD excluded. Cardiac MRI does not image coronary plaque or calcium. Encoders were fine-tuned on balanced case–control sets drawn from the training partition only; every risk analysis keeps the observed event rate. The polygenic risk score (PRS) was chosen on the training partition. Analyses are restricted to genetically defined White British participants.

Learned imaging representations outperform conventional phenotypes

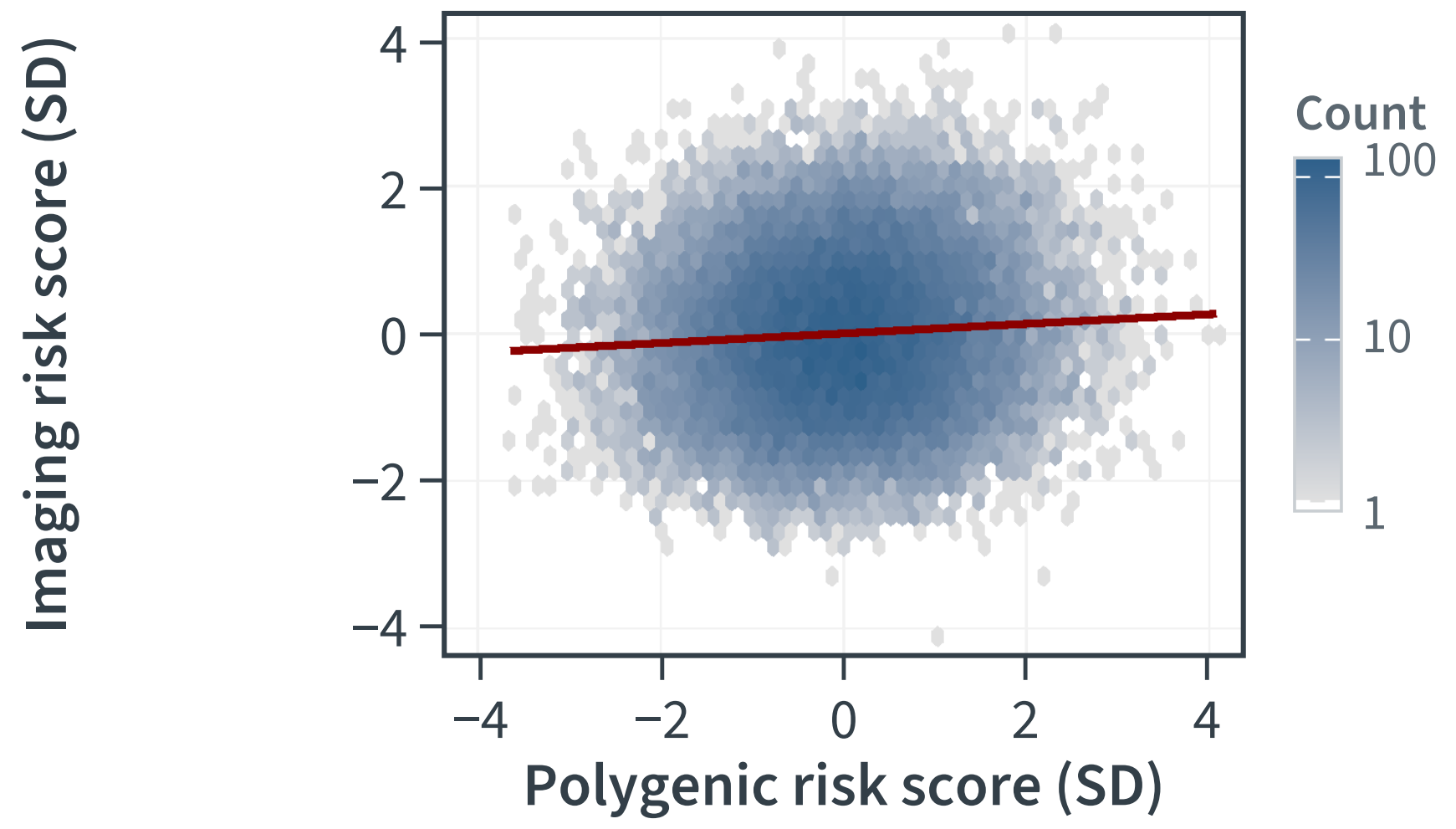


Reported representation benchmark (Fig. 2b), separate from the held-out Cox analyses. Leading canonical components of the embeddings align with stroke volume, ejection fraction and myocardial mass (first-component canonical correlations 0.79–0.84), and the embeddings retain additional predictive signal.

Imaging and genetics contribute complementary information

An imaging risk score (IRS; ridge-penalised Cox model on the 80 imaging principal components, five-fold cross-fitted) and the PRS both predict CAD, yet they barely overlap.

Weak PRS–imaging risk correlation; shared variance <1%



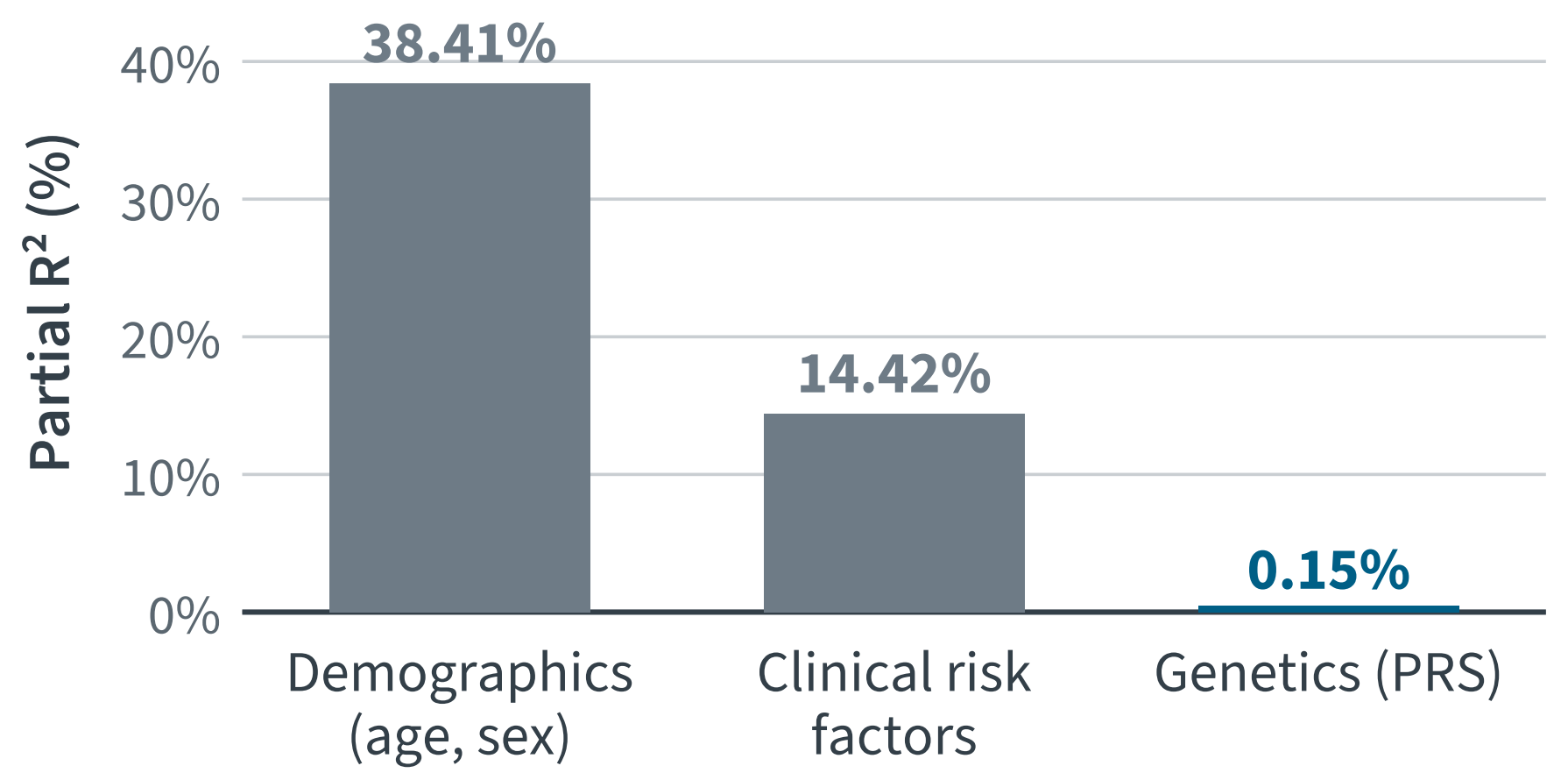
$r = 0.065$
Pearson correlation, PRS vs IRS (Fig. 3c)

$R^2 = 0.004$
shared variance (text summary: R^2 <1%)

$n = 29{,}075$
participants in the plotted training-partition data

PRS distribution is flat across imaging-risk quintiles (Fig. 3d). Weak overlap supports complementarity; it does not establish biological independence or show that imaging risk is wholly acquired.

What explains the imaging risk score? (Fig. 3e)



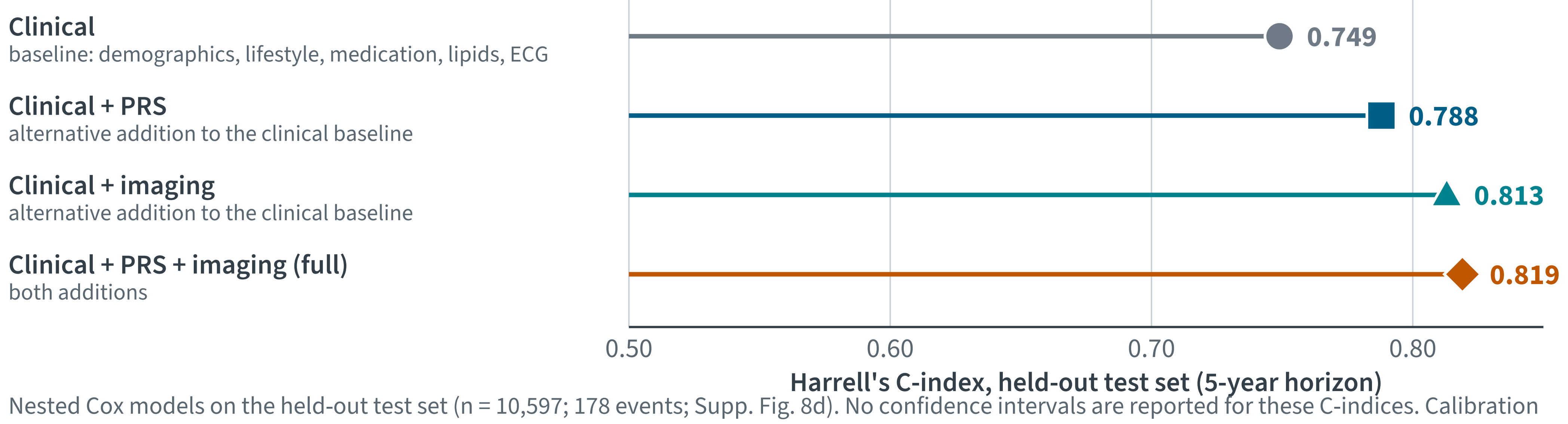
Which feature groups carry non-redundant information?

Largest loss of explained variance when removed
- Long-axis cardiac MRI
- Medication history
- Polygenic risk score
- Aortic distensibility MRI

Little loss after adjustment for clinical variables
- Liver MRI
- Pancreas MRI
- Whole-body DXA
- Blood-flow MRI

Qualitative grouping from the manuscript text (Fig. 3b, 3f); values not reproduced.

Held-out discrimination: full model 0.819 vs clinical 0.749 (absolute C-index difference 0.070)



Nested Cox models on the held-out test set (n = 10,597; 178 events; Supp. Fig. 8d). No confidence intervals are reported for these C-indices. Calibration slopes 0.99–1.08 across all four models.

Integrated risk stratification and robustness

Sequential stratification of the held-out set (Fig. 4e): each tier keeps the top third of the previous tier, ranked by one risk dimension at a time.

Tier 1
Top third by clinical-model risk
8.0%
5-year CAD incidence

→

Tier 2
Within Tier 1: top third by PRS
11.3%
5-year CAD incidence

→

Tier 3
Within Tier 2: top third by imaging risk score
15.9%
5-year CAD incidence

36.8% **Secondary subgroup:** within Tier 3, participants in the top quintile of *both* PRS and imaging risk (quintiles defined within the tier), 5-year Kaplan–Meier incidence. Subgroup size and confidence interval are not reported, so this small stratum needs cautious interpretation. Context: the observed event proportion in the full cohort is 1.6% (crude, not a 5-year estimate).

Leave-one-centre-out evaluation within UK Biobank (not external validation)

Held-out centre	Test cases / controls	AUC	C-index
Reading	71 / 7,607	0.805	0.762
Cheadle	409 / 21,942	0.822	0.748
Bristol	82 / 3,307	0.785	0.751
Newcastle	115 / 8,740	0.795	0.732

Site-analysis models trained on three centres and tested on the fourth (Supp. Table 13). The site feature set is not identical to the main full model, so these values are not comparable with the 0.819 above.

Calibration, reclassification and accuracy (held-out)

0.634
continuous NRI, full vs clinical (95% CI 0.534–0.734)

5.5% → 9.5%
index of prediction accuracy, clinical → full

0.99–1.08
calibration slopes, all nested models

Full-model net benefit was highest at 5–30% threshold probabilities (decision-curve analysis). Continuous NRI is not the share of people correctly reclassified, and net benefit is not demonstrated patient benefit.

Conclusion

Selected cardiac imaging (long-axis cine and aortic distensibility MRI) and genome-wide polygenic risk add partially non-overlapping information to clinical variables, improving held-out discrimination, calibration and enrichment of high-incidence subgroups in this cohort.

Limitations

- Selected, complete-case, White British UK Biobank sample; absolute risks may not transport to other settings.
- No fully external validation; site robustness is within one cohort.
- No coronary calcium or CT-angiography benchmark; cardiac MRI does not image plaque.
- Observational prediction: clinical utility, test ordering and cost-effectiveness remain untested.
- Extreme-risk strata are small and their precision is not reported.

Data and code
UK Biobank data are available to approved researchers through application (this work: application 65439); they are not public. Analysis and figure code: github.com/Devanshpandey/Multimodal-integration-for-coronary-artery-disease-risk-prediction (code only; no participant data or model weights). Genome-wide polygenic risk score; no individual gene-level discovery claims.

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